



SCRUTINY ACADEMY

DOCTOR IN THE MAKING SERIES



PERSONALISED NEET 2028 EXAMINATION

UNIT TEST 3

Question Paper

EXCLUSIVELY DESIGNED FOR

SANVI MANTRI

NEET 2028 Aspirant

180

QUESTIONS

720

MAXIMUM MARKS

3 Hours

DURATION

+4 / -1

MARKING

Test syllabus

Physics	Laws of Motion: friction through the end; Work, Energy & Power: up to Power	45 MCQs
Chemistry	Thermodynamics	45 MCQs
Botany	Morphology: Fabaceae, Solanaceae & Liliaceae; complete Anatomy of Flowering Plants	45 MCQs
Zoology	Structural Organisation in Animals: Frog only	45 MCQs

All the best, Sanvi!

You're one step closer to becoming a doctor. Stay calm, trust your preparation and give every question your honest best.

— Scrutiny Academy Team

EXAM DESIGNED BY SCRUTINY ACADEMY TEAM

READ CAREFULLY

General Instructions

1. The question paper contains **180 compulsory multiple-choice questions** divided equally among Physics, Chemistry, Botany and Zoology.
2. The total duration is **3 hours** and the maximum possible score is **720 marks**.
3. Each correct answer carries **+4 marks**. Each incorrect answer carries **–1 mark**. An unattempted question carries zero marks.
4. Each question has four alternatives—A, B, C and D—of which only one is the most appropriate answer.
5. Use the response sheet supplied at the end of this paper for rough marking, or submit answers through the official online test page.
6. Use $g = 10 \text{ m s}^{-2}$ wherever explicitly stated. Otherwise use the data provided in the question.
7. Tables and diagrams form an integral part of their respective questions.
8. Do not consult books, notes or online sources during the examination.

Question map

01–45

PHYSICS

46–90

CHEMISTRY

91–135

BOTANY

136–180

ZOOLOGY

Candidate details

NAME **Sanvi Mantri**

DATE _____

START TIME _____ END TIME _____

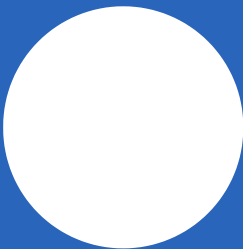
SIGNATURE

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Exam integrity note

The online test permits one attempt on the device where it is started. Begin only when ready to complete the full examination.



008. APPARENT WEIGHT

EASY

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A lift moves downward with constant speed. The apparent weight of a passenger of mass m is:

- ☐ A zero ☐ B mg
☐ C less than mg ☐ D greater than mg

009. IMPULSE AND MOMENTUM

MODERATE

A 0.2 kg ball moving at 10 m s^{-1} along $+x$ rebounds at 5 m s^{-1} along $-x$. The magnitude of impulse on the ball is:

- ☐ A 3 kg m s^{-1} ☐ B 2 kg m s^{-1}
☐ C 1 kg m s^{-1} ☐ D 5 kg m s^{-1}

010. IMPULSE AND MOMENTUM

MODERATE

A force–time graph is triangular, rising uniformly from 0 to 10 N in 2 s and falling uniformly to 0 in the next 2 s . The impulse is:

- ☐ A 10 N s ☐ B 30 N s
☐ C 20 N s ☐ D 40 N s

011. STATIC FRICTION

EASY

A 5 N horizontal force is applied to a block at rest. The limiting static friction is 8 N and the block does not move. The frictional force is:

- ☐ A 5 N ☐ B 3 N
☐ C 8 N ☐ D 0 N

012. LIMITING FRICTION

EASY

For a 10 kg block on a horizontal surface, $\mu_s = 0.4$ and $g = 10 \text{ m s}^{-2}$. The limiting friction is:

- ☐ A 4 N ☐ B 40 N
☐ C 25 N ☐ D 100 N

013. KINETIC FRICTION

MODERATE

A 20 N horizontal force moves a 5 kg block on a rough horizontal surface. If $\mu_k = 0.2$ and $g = 10 \text{ m s}^{-2}$, its acceleration is:

- ☐ A 4 m s^{-2} ☐ B 1 m s^{-2}
☐ C 3 m s^{-2} ☐ D 2 m s^{-2}

014. FRICTION WITH PULL

MODERATE

A 10 kg block is pulled by a 50 N force at 37° above the horizontal on a rough floor. If $\mu_k = 0.25$, $g = 10 \text{ m s}^{-2}$, $\sin 37^\circ = 0.6$ and $\cos 37^\circ = 0.8$, its acceleration is:

- ☐ A 2.25 m s^{-2} ☐ B 3.00 m s^{-2}
☐ C 1.50 m s^{-2} ☐ D 4.00 m s^{-2}

015. FRICTION WITH PUSH

MODERATE

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A 10 kg block is pushed by a 50 N force at 37° below the horizontal. For $\mu_k = 0.25$ and $g = 10 \text{ m s}^{-2}$, the kinetic friction is:

- ☐ A 17.5 N
- ☐ B 25.0 N
- ☐ C 37.5 N
- ☐ D 32.5 N

016. FRICTION ON INCLINE

EASY

A block is at rest on a rough incline and would slide downward if friction vanished. The static friction on it acts:

- ☐ A normal to the incline
- ☐ B down the incline
- ☐ C vertically upward
- ☐ D up the incline

017. ANGLE OF REPOSE

MODERATE

The coefficient of static friction between a block and an inclined plane is $1/\sqrt{3}$. The angle of repose is:

- ☐ A 30°
- ☐ B 45°
- ☐ C 60°
- ☐ D 90°

018. FRICTION ON CIRCULAR ROAD

MODERATE

A car moves on a level circular road of radius 20 m. If $\mu_s = 0.5$ and $g = 10 \text{ m s}^{-2}$, the maximum speed without skidding is:

- ☐ A 5 m s^{-1}
- ☐ B 20 m s^{-1}
- ☐ C 10 m s^{-1}
- ☐ D 15 m s^{-1}

019. APPLICATIONS OF FRICTION

EASY

While a person walks forward without slipping, the horizontal force exerted by the ground on the person is:

- ☐ A zero
- ☐ B backward static friction
- ☐ C forward kinetic friction
- ☐ D forward static friction

020. NEWTON'S THIRD LAW

EASY

Action and reaction forces do not cancel each other because they:

- ☐ A are unequal
- ☐ B act at different times
- ☐ C are not oppositely directed
- ☐ D act on different bodies

021. WORK BY CONSTANT FORCE

EASY

A 10 N force produces a displacement of 5 m at 60° to the force. Work done is:

- ☐ A 0 J
- ☐ B 100 J
- ☐ C 25 J
- ☐ D 50 J

A block moves horizontally on a fixed horizontal table. The work done by the table's normal reaction is:

- ☐ A positive
 ☐ B zero
- ☐ C negative
 ☐ D equal to mgs

For a body sliding on a rough horizontal surface, the work done by kinetic friction is generally:

- ☐ A positive
 ☐ B negative
- ☐ C zero
 ☐ D always infinite

The kinetic energy of a 2 kg object moving at 3 m s^{-1} is:

- ☐ A 3 J
 ☐ B 9 J
- ☐ C 18 J
 ☐ D 6 J

A particle initially has 10 J kinetic energy. If the net work done on it is 40 J, its final kinetic energy is:

- ☐ A 40 J
 ☐ B 50 J
- ☐ C 30 J
 ☐ D 400 J

Two particles have equal momentum. Their masses are m and $2m$. The ratio of their kinetic energies $K_m : K_{2m}$ is:

- ☐ A 1:1
 ☐ B 1:2
- ☐ C 2:1
 ☐ D 4:1

The force along x varies as shown:

$x \text{ (m)}$	0	2	3
$F \text{ (N)}$	0	10	10

From 0–2 m, F rises linearly; from 2–3 m it is constant. Work done from 0 to 3 m is:

- ☐ A 10 J
 ☐ B 15 J
- ☐ C 20 J
 ☐ D 30 J

A spring of force constant 200 N m^{-1} is compressed by 0.10 m. Energy stored in it is:

- ☐ A 1.0 J
 ☐ B 2.0 J
- ☐ C 0.5 J
 ☐ D 10 J

029. WORK BY SPRING

MODERATE

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A spring with $k = 100 \text{ N m}^{-1}$ is released from compression $x = 0.20 \text{ m}$ to its natural length. Work done by the spring is:

- (A) 4 J (B) 0 J
(C) -2 J (D) 2 J

030. GRAVITATIONAL POTENTIAL ENERGY

EASY

A 2 kg object is raised vertically through 5 m. Taking $g = 10 \text{ m s}^{-2}$, the increase in gravitational potential energy is:

- (A) 100 J (B) 10 J
(C) 50 J (D) 25 J

031. CONSERVATIVE FORCES

EASY

The work done by a conservative force over any closed path is:

- (A) negative (B) zero
(C) positive (D) path dependent

032. MECHANICAL ENERGY CONSERVATION

MODERATE

A body starts from rest and slides down a smooth track through a vertical height of 5 m. For $g = 10 \text{ m s}^{-2}$, its speed at the bottom is:

- (A) 10 m s^{-1} (B) 25 m s^{-1}
(C) 5 m s^{-1} (D) 50 m s^{-1}

033. MECHANICAL ENERGY WITH FRICTION

MODERATE

A 2 kg body starts from rest and descends through 5 m. Friction dissipates 20 J. Take $g = 10 \text{ m s}^{-2}$. Its speed at the bottom is closest to:

- (A) 6.3 m s^{-1} (B) 12.6 m s^{-1}
(C) 8.9 m s^{-1} (D) 10.0 m s^{-1}

034. MECHANICAL ENERGY CONSERVATION

EASY

For a simple pendulum oscillating without air resistance, which quantity remains constant?

- (A) Mechanical energy (B) Potential energy only
(C) Kinetic energy only (D) Speed

035. INSTANTANEOUS POWER

MODERATE

A 100 N force acts on an object moving at 5 m s^{-1} . The angle between force and velocity is 60° . Instantaneous power is:

- (A) 100 W (B) 500 W
(C) 250 W (D) 1000 W

036. AVERAGE POWER

EASY

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A 60 kg student climbs a vertical height of 10 m in 20 s. Take $g = 10 \text{ m s}^{-2}$. Average power developed is:

- ☐ A 300 W ☐ B 600 W
☐ C 30 W ☐ D 3000 W

037. POWER OF PUMP

MODERATE

A pump lifts 100 kg of water per minute to a height of 12 m. Take $g = 10 \text{ m s}^{-2}$. Its useful power is:

- ☐ A 200 W ☐ B 120 W
☐ C 1200 W ☐ D 20 W

038. POWER AND EFFICIENCY

MODERATE

A motor takes 2.0 kW electrical power and delivers 1.5 kW mechanical power. Its efficiency is:

- ☐ A 133% ☐ B 50%
☐ C 25% ☐ D 75%

039. POWER AT CONSTANT SPEED

MODERATE

A car moves at a constant 20 m s^{-1} against a total resistive force of 800 N. Engine power delivered at the wheels is:

- ☐ A 4 kW ☐ B 8 kW
☐ C 40 kW ☐ D 16 kW

040. INSTANTANEOUS POWER

EASY

A constant force remains parallel to a body's velocity. If the speed doubles, instantaneous power due to the force becomes:

- ☐ A half ☐ B unchanged
☐ C four times ☐ D double

041. VARIABLE FORCE WORK

MODERATE

A one-dimensional force is $F(x) = 2x \text{ N}$, where x is in metres. Work done from $x = 0$ to $x = 3 \text{ m}$ is:

- ☐ A 3 J ☐ B 6 J
☐ C 18 J ☐ D 9 J

042. POTENTIAL ENERGY

MODERATE

The potential energy of a particle is $U(x) = 3x^2 \text{ J}$. The force at $x = 2 \text{ m}$ is:

- ☐ A 12 N ☐ B -6 N
☐ C -12 N ☐ D 6 N

043.

FRICTION DATA INTERPRETATION

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MODERATE
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A 4 kg block rests on a horizontal surface where $\mu_s = 0.50$ and $\mu_k = 0.30$. Take $g = 10 \text{ m s}^{-2}$.

Applied force	Motion	Friction magnitude
12 N	At rest	?
24 N	Sliding	?

The two friction values are respectively:

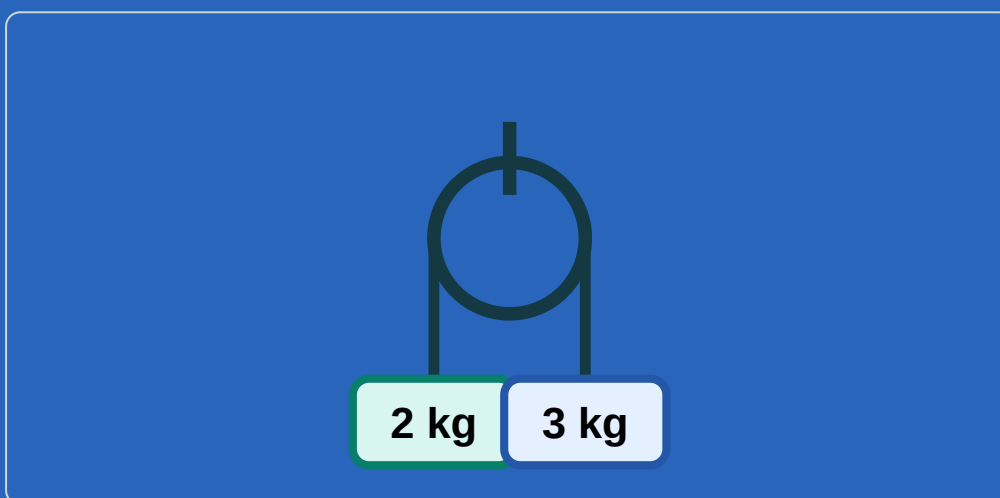
- ☐ A 12 N and 12 N ☐ B 20 N and 12 N
☐ C 12 N and 20 N ☐ D 20 N and 20 N

044.

PULLEY SYSTEM

MODERATE

The light string and ideal pulley connect 2 kg and 3 kg blocks as shown. Take $g = 10 \text{ m s}^{-2}$.



The magnitude of acceleration and string tension are:

- ☐ A 4 m s^{-2} and 12 N ☐ B 2 m s^{-2} and 24 N
☐ C 2 m s^{-2} and 30 N ☐ D 5 m s^{-2} and 20 N

045.

CONCEPT INTEGRATION

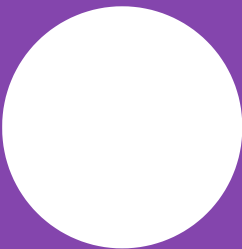
MODERATE

Consider the statements:

- I. If only conservative forces do work, total mechanical energy remains constant.
 II. Work done by kinetic friction generally depends on the path length.

Choose the correct option.

- ☐ A I is true but II is false ☐ B Both I and II are false
☐ C Both I and II are true ☐ D I is false but II is true



In thermodynamics, the part of the universe selected for study is called the:

- ☐ A surroundings
- ☐ B boundary
- ☐ C system
- ☐ D reservoir

Which system can exchange both matter and energy with its surroundings?

- ☐ A Isolated system
- ☐ B Closed system
- ☐ C Adiabatic closed system
- ☐ D Open system

A closed system may exchange with its surroundings:

- ☐ A matter only
- ☐ B energy only
- ☐ C both matter and energy
- ☐ D neither matter nor energy

An ideal thermos flask is commonly approximated as a/an:

- ☐ A isolated system
- ☐ B closed isothermal system
- ☐ C open system
- ☐ D homogeneous open system

Which of the following is a state function?

- ☐ A Heat q
- ☐ B Work w
- ☐ C Internal energy U
- ☐ D Distance travelled

Which property is extensive?

- ☐ A Enthalpy
- ☐ B Pressure
- ☐ C Temperature
- ☐ D Density

Which property is intensive?

- ☐ A Volume
- ☐ B Internal energy
- ☐ C Number of moles
- ☐ D Temperature

053.	SIGN CONVENTION	EASY
SCRUTINY: JEE ADVANCED PRACTICE EXERCISE: PAPER-1 According to the chemistry sign convention, heat absorbed by a system has: SANVI MANTRI		
A a sign determined only by volume B $q < 0$ C $q = 0$ D $q > 0$		
054.	PRESSURE–VOLUME WORK	EASY
A gas expands against a constant external pressure. The pressure–volume work w done on the system is:		
A negative B always equal to q C zero D positive		
055.	FIRST LAW	EASY
Using the chemistry sign convention, the first law of thermodynamics is:		
A $\Delta U = q - w$ B $\Delta U = q + w$ C $\Delta U = w - q$ D $\Delta U = -q - w$		
056.	FIRST LAW	MODERATE
A system absorbs 50 kJ of heat and does 20 kJ of work on the surroundings. Its change in internal energy is:		
A –70 kJ B +30 kJ C –30 kJ D +70 kJ		
057.	ADIABATIC PROCESS	EASY
For an adiabatic process:		
A $q = 0$ B $w = 0$ C $\Delta U = 0$ D $\Delta H = 0$		
058.	CYCLIC PROCESS	EASY
After a system completes a thermodynamic cycle and returns to its initial state, its ΔU is:		
A zero B negative C equal to total path length D positive		
059.	CALORIMETRY	EASY
At constant volume, when only pressure–volume work is possible, heat exchanged is equal to:		
A ΔH B ΔU C $-\Delta U$ D ΔG		
060.	ENTHALPY	EASY
At constant pressure and with only pressure–volume work, the heat exchanged equals:		
A ΔU B ΔH C ΔS D $-\Delta H$		

061.

 ΔH AND ΔU

MODERATE

SCRUPTURE MCQ TEST 2023 UNIT TEST 1

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For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ at temperature T , the relation between ΔH and ΔU is:

- ☐ A $\Delta H = \Delta U + 2RT$ ☐ B $\Delta H = \Delta U$
☐ C $\Delta H = \Delta U - 2RT$ ☐ D $\Delta H = \Delta U + 4RT$

062.

HEAT CAPACITY

MODERATE

Two moles of a substance with molar heat capacity $25 \text{ J mol}^{-1} \text{ K}^{-1}$ are heated through 20 K without a phase change. Heat supplied is:

- ☐ A 250 J ☐ B 1000 J
☐ C 2000 J ☐ D 500 J

063.

CALORIMETRY

EASY

How much heat is required to raise the temperature of 100 g water by 5 K ? Take specific heat $= 4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

- ☐ A 84 J ☐ B 840 J
☐ C 2100 J ☐ D 420 J

064.

CALORIMETRY

EASY

A bomb calorimeter operates essentially at constant volume and directly determines:

- ☐ A ΔU of combustion ☐ B ΔH of combustion
☐ C ΔG of combustion ☐ D entropy of combustion

065.

CALORIMETRY

EASY

A coffee-cup calorimeter open to the atmosphere most directly measures heat change at:

- ☐ A constant volume ☐ B constant pressure
☐ C constant entropy ☐ D zero temperature

066.

REACTION ENTHALPY

EASY

For an exothermic reaction at constant pressure:

- ☐ A $\Delta H = 0$ ☐ B $\Delta H > 0$
☐ C $\Delta H < 0$ ☐ D ΔH must equal ΔU

067.

ENTHALPY DIAGRAM

EASY

In an endothermic reaction, the products generally lie:

- ☐ A at lower enthalpy than reactants ☐ B at zero enthalpy
☐ C at the same enthalpy as reactants ☐ D at higher enthalpy than reactants

Hess's law is valid primarily because enthalpy is a:

- ☐ A path function
- ☐ B dimensionless quantity
- ☐ C state function
- ☐ D vector quantity

The standard enthalpy of formation of $O_2(g)$ in its standard state is:

- ☐ A 0 kJ mol^{-1}
- ☐ B $+1 \text{ kJ mol}^{-1}$
- ☐ C -1 kJ mol^{-1}
- ☐ D temperature dependent but never zero

Standard enthalpy of formation is the enthalpy change when:

- ☐ A one mole of solid changes to gas
- ☐ B one mole of any substance burns completely
- ☐ C an acid and a base form one mole of salt
- ☐ D one mole of a compound forms from its elements in their standard states

The standard enthalpy of combustion of methane is -890 kJ mol^{-1} . Heat released by complete combustion of 0.50 mol CH_4 is:

- ☐ A 1780 kJ
- ☐ B 222.5 kJ
- ☐ C 890 kJ
- ☐ D 445 kJ

Use the data below for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$.

Substance	$\Delta_f H^\circ \text{ (kJ mol}^{-1}\text{)}$
$CH_4(g)$	-75
$CO_2(g)$	-394
$H_2O(l)$	-286
$O_2(g)$	0

The reaction enthalpy is:

- ☐ A -605 kJ mol^{-1}
- ☐ B $+967 \text{ kJ mol}^{-1}$
- ☐ C $+891 \text{ kJ mol}^{-1}$
- ☐ D -891 kJ mol^{-1}

Given bond enthalpies $H-H = 436$, $Cl-Cl = 243$ and $H-Cl = 431 \text{ kJ mol}^{-1}$, estimate ΔH for $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$.

- ☐ A -674 kJ mol^{-1}
- ☐ B $+183 \text{ kJ mol}^{-1}$
- ☐ C $+862 \text{ kJ mol}^{-1}$
- ☐ D -183 kJ mol^{-1}

Which change represents atomisation of carbon in its standard state?

- ☐ A $\text{C(g)} \rightarrow \text{C(graphite)}$
☐ B $\text{C(graphite)} \rightarrow \text{C(g)}$
- ☐ C $\text{CO}_2\text{(g)} \rightarrow \text{C(graphite)} + \text{O}_2\text{(g)}$
☐ D $\text{C(diamond)} \rightarrow \text{C(graphite)}$

For a substance below its critical temperature, the enthalpy change for sublimation (solid $\rightarrow$ gas) is normally:

- ☐ A zero
 ☐ B negative
- ☐ C positive
 ☐ D always smaller than fusion enthalpy

For the same pure substance at the same temperature, the usual order of molar entropy is:

- ☐ A $S(\text{solid}) > S(\text{liquid}) > S(\text{gas})$
☐ B $S(\text{solid}) = S(\text{liquid}) = S(\text{gas})$
- ☐ C $S(\text{liquid}) > S(\text{gas}) > S(\text{solid})$
☐ D $S(\text{gas}) > S(\text{liquid}) > S(\text{solid})$

Spontaneous mixing of two different ideal gases at the same temperature generally causes entropy to:

- ☐ A become undefined
 ☐ B increase
- ☐ C remain zero
 ☐ D decrease

For a spontaneous process in an isolated system, the entropy change of the universe is:

- ☐ A negative
 ☐ B zero only
- ☐ C positive
 ☐ D equal to $\Delta H/T$ in every case

For an ideal reversible phase transition occurring at equilibrium, $\Delta S(\text{universe})$ is:

- ☐ A less than zero
 ☐ B equal to ΔH of the system
- ☐ C greater than zero
 ☐ D equal to zero

A system releases 20 kJ of heat to surroundings maintained at 300 K. The entropy change of the surroundings is approximately:

- ☐ A -66.7 J K^{-1}
☐ B $+66.7 \text{ J K}^{-1}$
- ☐ C -0.067 J K^{-1}
☐ D $+0.067 \text{ J K}^{-1}$

The Gibbs energy change at constant temperature is given by:

- ☐ A $\Delta G = \Delta H - T\Delta S$
☐ B $\Delta G = \Delta H + T\Delta S$
- ☐ C $\Delta G = T\Delta H - \Delta S$
☐ D $\Delta G = \Delta U + P\Delta V$ only

At constant temperature and pressure, a process is thermodynamically spontaneous when:

- ☐ A $\Delta H = 0$
☐ B $\Delta G > 0$
- ☐ C $\Delta G = 0$ only
 ☐ D $\Delta G < 0$

At equilibrium under constant temperature and pressure:

- ☐ A $\Delta G < 0$
☐ B $\Delta G > 0$
- ☐ C $\Delta G = 0$
☐ D $\Delta S(\text{system}) = 0$ necessarily

A reaction has $\Delta H < 0$ and $\Delta S > 0$. It is expected to be spontaneous:

- ☐ A only at low temperatures
 ☐ B at all temperatures
- ☐ C only at high temperatures
 ☐ D at no temperature

For a reaction, $\Delta H = +40 \text{ kJ mol}^{-1}$ and $\Delta S = +100 \text{ J mol}^{-1} \text{ K}^{-1}$, assumed temperature independent. Above what temperature does ΔG become negative?

- ☐ A 400 K
 ☐ B 200 K
- ☐ C 100 K
 ☐ D 800 K

A reaction with $\Delta H < 0$ and $\Delta S < 0$ is generally spontaneous:

- ☐ A at all temperatures
 ☐ B at high temperature
- ☐ C at low temperature
 ☐ D at no temperature

For a reaction at a specified temperature, $K > 1$. From $\Delta G^\circ = -RT \ln K$, one can conclude that:

- ☐ A ΔG° is positive
 ☐ B ΔG° is negative
- ☐ C ΔG° is zero
 ☐ D ΔG° equals K

088.

SCRU

An ideal gas undergoes isothermal expansion. Its change in internal energy is:

089.

Between the same initial and final states, isothermal reversible expansion of an ideal gas gives:

- ☐ A maximum magnitude of expansion work
☐ B minimum magnitude of expansion work
☐ C zero work
☐ D work independent of path

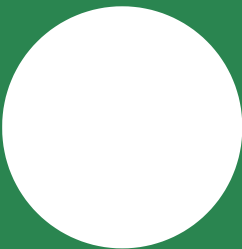
090.

A gas undergoes the three processes listed below using the chemistry sign convention.

Process	q (kJ)	w (kJ)
A	+30	−10
B	−5	+15
C	0	−8

Which set of ΔU values (A, B, C) is correct?

- (A) +20, -20, -8 kJ
- (B) +40, -20, +8 kJ
- (C) +20, +10, -8 kJ
- (D) -20, +10, +8 kJ



091.

FABACEAE

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EASY

A flower is zygomorphic and papilionaceous, with ten stamens usually arranged as (9)+1. It belongs to:

- ☐ A Solanaceae
- ☐ B Fabaceae
- ☐ C Liliaceae
- ☐ D Brassicaceae

092.

FABACEAE

EASY

In vexillary aestivation of Fabaceae:

- ☐ A the keel petal overlaps all other petals
- ☐ B each petal overlaps the next in one direction
- ☐ C all five petals meet but do not overlap
- ☐ D the largest standard petal overlaps the two wings

093.

FABACEAE

EASY

The typical androecium of pea is:

- ☐ A numerous monadelphous stamens
- ☐ B 6 epiphyllous stamens
- ☐ C 5 epipetalous stamens
- ☐ D 10 diadelphous stamens (9)+1

094.

FABACEAE

MODERATE

The Fabaceae gynoecium is usually:

- ☐ A bicarpellary, syncarpous with axile placentation
- ☐ B monocarpellary with basal placentation and inferior ovary
- ☐ C tricarpeal with an inferior ovary
- ☐ D monocarpellary with a superior ovary and marginal placentation

095.

FABACEAE

EASY

Which fruit–seed combination is typical of Fabaceae?

- ☐ A Capsule; winged seeds
- ☐ B Legume; generally non-endospermic seeds
- ☐ C Berry; endospermic seeds
- ☐ D Drupe; one endospermic seed

096.

SOLANACEAE

EASY

A typical Solanaceae flower is:

- ☐ A zygomorphic, unisexual and trimerous
- ☐ B actinomorphic, bisexual and pentamerous
- ☐ C actinomorphic, unisexual and tetramerous
- ☐ D zygomorphic, bisexual and trimerous

097.

SOLANACEAE

EASY

The five stamens of Solanaceae are usually:

- ☐ A epipetalous and alternating with petals
- ☐ B diadelphous and ten in number
- ☐ C epiphyllous and three in number
- ☐ D free and opposite the petals

The ovary in Solanaceae is typically:

- ☐ A

superior, bicarpellary, syncarpous with axile placentation
- ☐ B

inferior, tricarpellary with parietal placentation
- ☐ C

superior, monocarpellary with marginal placentation
- ☐ D

half-inferior and unilocular with basal placentation

Which pair correctly represents common fruit types in Solanaceae?

- ☐ A

Legume and lomentum
- ☐ B

Caryopsis and achene
- ☐ C

Berry and capsule
- ☐ D

Drupe and samara

Study the comparison table.

Feature	Family X
Symmetry	Actinomorphic
Stamens	5, epipetalous
Ovary	Superior, bicarpellary
Fruit	Berry or capsule

Family X is:

- ☐ A

Solanaceae
- ☐ B

Fabaceae
- ☐ C

Liliaceae
- ☐ D

Poaceae

In Liliaceae, the floral envelope is commonly described as:

- ☐ A

calyx 5 plus corolla 5
- ☐ B

a papilionaceous corolla
- ☐ C

four sepals and four petals
- ☐ D

six petaloid tepals arranged 3+3

The androecium of a typical Liliaceae flower consists of:

- ☐ A

5 epipetalous stamens
- ☐ B

10 diadelphous stamens
- ☐ C

6 epiphyllous stamens
- ☐ D

numerous free stamens

The gynoecium of Liliaceae is generally:

- ☐ A

monocarpellary, unilocular and inferior
- ☐ B

tricarpellary, syncarpous, trilocular and superior
- ☐ C

bicarpellary, syncarpous and inferior
- ☐ D

five-carpellary with marginal placentation

104.

LILIACEAE

EASY

SCRUTINY ACADEMY • NEET 2028 UNIT TEST 3

SANVI MANTRI

Which vegetative combination commonly occurs in Liliaceae?

- ☐ A Fibrous root and parallel venation
 ☐ B Pneumatophores and net venation
 ☐ C Prop roots and dichotomous venation
 ☐ D Tap root and reticulate venation

105.

PLANT FAMILIES

MODERATE

Which row is entirely correct?

Option	Family	Androecium	Placentation	Typical seed
A	Fabaceae	10, (9)+1	Marginal	Non-endospermic
B	Solanaceae	6, epiphyllous	Marginal	Non-endospermic
C	Liliaceae	5, epipetalous	Parietal	Non-endospermic
D	Fabaceae	5, epipetalous	Axile	Endospermic

- ☐ A Row B
 ☐ B Row A
 ☐ C Row C
 ☐ D Row D

106.

MERISTEMATIC TISSUES

EASY

Meristematic cells generally possess:

- ☐ A thick lignified walls and large intercellular spaces
 ☐ B large vacuoles and no nuclei
 ☐ C dead protoplasts and suberised walls
 ☐ D dense cytoplasm, prominent nuclei and active division

107.

MERISTEMATIC TISSUES

EASY

The meristem mainly responsible for increase in girth of stems and roots is:

- ☐ A apical meristem
 ☐ B intercalary meristem
 ☐ C protoderm
 ☐ D lateral meristem

108.

MERISTEMATIC TISSUES

EASY

Intercalary meristem is commonly located:

- ☐ A at root cap only
 ☐ B inside vessel elements
 ☐ C between mature tissues, often near internode or leaf bases
 ☐ D only in the bark of woody dicots

109.

SIMPLE PERMANENT TISSUES

EASY

Which tissue consists of living, thin-walled cells and often performs storage and photosynthesis?

- ☐ A Parenchyma
 ☐ B Sclerenchyma
 ☐ C Xylem fibres
 ☐ D Cork

Uneven thickening at the cell corners is characteristic of:

- ☐ A parenchyma
 ☐ B vessel elements
- ☐ C sclereids
 ☐ D collenchyma

Mature sclerenchyma cells are usually:

- ☐ A living and chlorophyllous
 ☐ B thin-walled and highly vacuolated
- ☐ C dead with thick lignified walls
 ☐ D living with corner thickening only

Among xylem elements, the only one that is living at maturity is:

- ☐ A tracheid
 ☐ B xylem parenchyma
- ☐ C xylem fibre
 ☐ D vessel element

Which phloem element is generally dead at maturity?

- ☐ A Phloem fibre
 ☐ B Companion cell
- ☐ C Sieve-tube element
 ☐ D Phloem parenchyma

The stomatal apparatus includes:

- ☐ A a stoma and only xylem cells
 ☐ B a lenticel and cork cells
- ☐ C one guard cell and one trichome
 ☐ D a stomatal pore, two guard cells and associated subsidiary cells

In grasses, guard cells are typically:

- ☐ A kidney-shaped
 ☐ B spherical and without chloroplasts
- ☐ C triangular
 ☐ D dumb-bell shaped

All tissues inside the epidermis except the vascular bundles constitute the:

- ☐ A epidermal tissue system
 ☐ B ground tissue system
- ☐ C vascular tissue system
 ☐ D periderm only

117. VASCULAR BUNDLES

MODERATE

SCRUTINY ACADEMY • NEET 2020 UNIT TEST 3

SANVI MANTRI

A vascular bundle in which xylem and phloem occur on different radii is called:

- ☐ A conjoint
- ☐ B collateral closed
- ☐ C radial
- ☐ D bicollateral

118. DICOT ROOT

EASY

The primary xylem of a typical dicot root is:

- ☐ A endarch
- ☐ B concentric
- ☐ C exarch
- ☐ D mesarch

119. ROOT ANATOMY

MODERATE

Compared with a dicot root, a typical monocot root usually has:

- ☐ A fewer than four xylem bundles and no pith
- ☐ B conjoint open bundles in a ring
- ☐ C many xylem bundles and a large pith
- ☐ D endarch xylem and secondary growth

120. DICOT ROOT

MODERATE

The innermost cortical layer of a root, with Casparian strips in its radial and tangential walls, is the:

- ☐ A epiblema
- ☐ B endodermis
- ☐ C pericycle
- ☐ D pith

121. DICOT STEM

EASY

Vascular bundles in a young dicot stem are usually:

- ☐ A radial and exarch
- ☐ B concentric and amphicribal
- ☐ C conjoint, collateral, open and arranged in a ring
- ☐ D conjoint, collateral, closed and scattered

122. MONOCOT STEM

EASY

A typical monocot stem has vascular bundles that are:

- ☐ A arranged in a ring and open
- ☐ B restricted to the pith
- ☐ C radial and exarch
- ☐ D scattered in ground tissue and closed

123. MONOCOT STEM

MODERATE

The water-containing cavity commonly seen within a monocot stem vascular bundle is the:

- ☐ A resin canal
- ☐ B protoxylem lacuna
- ☐ C lenticel
- ☐ D secretory duct

Which comparison is correct?

Feature	Dicot stem	Monocot stem
I	Bundles in ring	Bundles scattered
II	Bundles open	Bundles closed
III	Hypodermis usually collenchymatous	Hypodermis usually sclerenchymatous

- ☐ A I only
 ☐ B II and III only
 ☐ C I, II and III
 ☐ D I and II only

125. DICOT LEAF

EASY

In a dorsiventral dicot leaf, mesophyll is differentiated into:

- ☐ A epidermis and pericycle
 ☐ B palisade and spongy parenchyma
 ☐ C xylem and phloem
 ☐ D bulliform and bundle-sheath cells only

126. MONOCOT LEAF

EASY

Bulliform cells of a grass leaf occur mainly in the:

- ☐ A upper epidermis
 ☐ B lower epidermis
 ☐ C vascular cambium
 ☐ D bundle phloem

127. LEAF VASCULAR BUNDLE

MODERATE

In a leaf vascular bundle, xylem generally faces the:

- ☐ A adaxial (upper) surface
 ☐ B abaxial (lower) surface
 ☐ C stomatal pore
 ☐ D bundle sheath equally on all sides

128. LEAF ANATOMY

MODERATE

Which feature most strongly identifies an isobilateral monocot leaf?

- ☐ A Undifferentiated mesophyll and stomata on both surfaces
 ☐ B Stomata mainly restricted to lower surface
 ☐ C Palisade tissue only below the upper epidermis
 ☐ D Radial vascular bundles with exarch xylem

129. SECONDARY GROWTH

EASY

During normal secondary growth in a dicot stem, vascular cambium produces:

- ☐ A secondary phloem inward and secondary xylem outward
 ☐ B secondary xylem inward and secondary phloem outward
 ☐ C cork outward and phelloderm inward
 ☐ D primary xylem and primary phloem only

130. VASCULAR CAMBIUM

MODERATE

SCRUTINY ACADEMY • NEET 2023 UNIT TEST 3

SANVI MANTRI

The continuous cambial ring of a dicot stem is formed by union of:

- ☐ A apical and intercalary meristems
- ☐ B primary xylem and medullary ray
- ☐ C intrafascicular and interfascicular cambia
- ☐ D cork cambium and pericycle

131. ANNUAL RINGS

MODERATE

Spring wood generally has:

- ☐ A larger vessel elements and lower density than autumn wood
- ☐ B narrower vessels and higher density than autumn wood
- ☐ C only phloem and no xylem
- ☐ D dark, compact cells with no conduction

132. HEARTWOOD AND SAPWOOD

EASY

The physiologically active, lighter-coloured outer secondary xylem that conducts water is:

- ☐ A sapwood
- ☐ B heartwood
- ☐ C phellem
- ☐ D phelloderm

133. CORK CAMBIUM

MODERATE

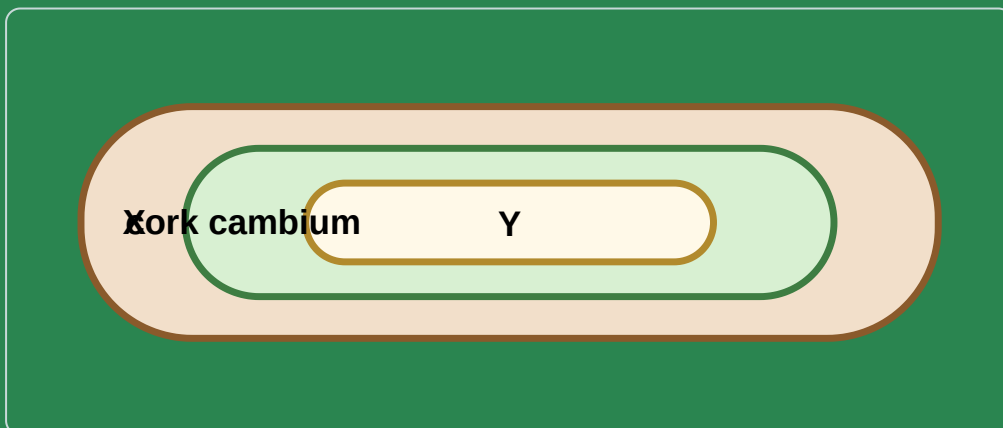
Cork cambium usually cuts off:

- ☐ A phellem inward and phelloderm outward
- ☐ B secondary xylem outward only
- ☐ C phellem outward and phelloderm inward
- ☐ D companion cells on both sides

134. SECONDARY GROWTH DIAGRAM

MODERATE

The simplified stem sector shows tissue X produced outside and tissue Y produced inside by the same lateral meristem.

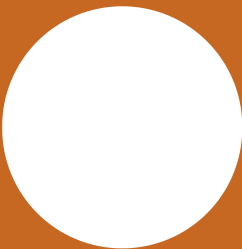


X and Y are respectively:

- ☐ A secondary xylem and secondary phloem
- ☐ B phelloderm and phellem
- ☐ C phellem and phelloderm
- ☐ D epidermis and pericycle

Which statement about lenticels is correct?

- ☐ A They are permanently closed stomata of secondary xylem
- ☐ B They produce annual rings
- ☐ C They transport sugars through heartwood
- ☐ D They permit gaseous exchange through the periderm



The common frog is described as amphibious because it:

- ☐ A lives only in deep water
- ☐ B can live both on land and in fresh water
- ☐ C flies between ponds
- ☐ D has gills throughout adult life

137. EXTERNAL MORPHOLOGY

EASY

The adult frog's body is divided mainly into:

- ☐ A head, neck and trunk
- ☐ B cephalothorax and abdomen
- ☐ C head and trunk
- ☐ D head, thorax and tail

138. EXTERNAL MORPHOLOGY

EASY

The frog's skin is kept moist and slippery mainly by:

- ☐ A sebaceous glands
- ☐ B salivary glands
- ☐ C sweat glands
- ☐ D mucous glands

139. EXTERNAL MORPHOLOGY

EASY

Frogs normally absorb water primarily through:

- ☐ A the mouth while drinking
- ☐ B the tympanum
- ☐ C the lungs
- ☐ D the skin

140. PROTECTIVE COLOURATION

EASY

The darker dorsal colour and lighter ventral surface of a frog mainly aid in:

- ☐ A digestion
- ☐ B camouflage
- ☐ C excretion
- ☐ D sound production

141. SEASONAL BEHAVIOUR

EASY

The summer sleep and winter sleep of frogs are respectively called:

- ☐ A diapause and migration
- ☐ B hibernation and aestivation
- ☐ C metamorphosis and moulting
- ☐ D aestivation and hibernation

142. SENSE ORGANS

EASY

The transparent membrane that protects a frog's eye in water is the:

- ☐ A tympanum
- ☐ B cornea
- ☐ C iris
- ☐ D nictitating membrane

143. SENSE ORGANS

EASY

SCRUTINISE QUESTIONS FIRST, THEN UNIT TOPICS

SANVI MANTRI

The circular tympanum visible behind each eye represents the:

- ☐ A poison gland
- ☐ B external nostril
- ☐ C vocal sac
- ☐ D eardrum

144. LIMBS

EASY

Webbing between the toes of a frog's hind limbs is chiefly an adaptation for:

- ☐ A swimming
- ☐ B burrowing
- ☐ C capturing insects
- ☐ D hearing

145. SEXUAL DIMORPHISM

EASY

A male frog can commonly be distinguished from a female by the presence of:

- ☐ A paired vocal sacs and copulatory pads
- ☐ B webbed hind feet only
- ☐ C a cloaca
- ☐ D two eyes and a tympanum

146. EXTERNAL MORPHOLOGY

MODERATE

Which comparison is correct?

Character	Male frog	Female frog
I	Vocal sacs present	Vocal sacs absent
II	Copulatory pad present	Copulatory pad absent
III	Generally smaller	Generally larger

- ☐ A I only
- ☐ B II and III only
- ☐ C I, II and III
- ☐ D I and II only

147. DIGESTIVE SYSTEM

EASY

The common chamber into which the alimentary, urinary and reproductive tracts open is the:

- ☐ A pharynx
- ☐ B duodenum
- ☐ C rectum
- ☐ D cloaca

148. BUCCAL CAVITY

EASY

Teeth in a frog are primarily used to:

- ☐ A chew food thoroughly
- ☐ B grind plant fibre
- ☐ C inject venom
- ☐ D hold captured prey

149.

FEEDING

MODERATE

SCRUTINIZE SCORING: SCPT 1000 UNIT TEST 2

SCPT 1000 UNIT TEST 2

The frog's tongue is attached at its anterior end and is free and bifid posteriorly. This arrangement helps it to:

- ☐ A filter water
- ☐ B pump blood
- ☐ C capture prey rapidly
- ☐ D produce urine

150.

DIGESTIVE TRACT

EASY

The two main regions of a frog's small intestine are:

- ☐ A duodenum and ileum
- ☐ B ileum and colon
- ☐ C jejunum and rectum
- ☐ D oesophagus and duodenum

151.

DIGESTIVE GLANDS

EASY

The bile-producing organ in a frog is the:

- ☐ A pancreas
- ☐ B stomach
- ☐ C spleen
- ☐ D liver

152.

DIGESTIVE GLANDS

MODERATE

The pancreas of a frog lies mainly:

- ☐ A within the urinary bladder
- ☐ B inside the heart
- ☐ C between the stomach and duodenum
- ☐ D on the dorsal surface of the brain

153.

DIGESTION

EASY

Protein digestion in a frog begins mainly in the:

- ☐ A mouth
- ☐ B cloaca
- ☐ C large intestine
- ☐ D stomach

154.

ABSORPTION

EASY

Most absorption of digested nutrients in a frog occurs through the inner lining of the:

- ☐ A small intestine
- ☐ B rectum
- ☐ C oesophagus
- ☐ D cloaca

155.

RESPIRATION

EASY

A frog submerged in water respire mainly through its:

- ☐ A buccal cavity only
- ☐ B lungs
- ☐ C moist skin
- ☐ D external gills

156.

RESPIRATION

EASY

On land, an adult frog may exchange gases through:

- ☐ A lungs, skin and buccopharyngeal lining
- ☐ B gills only
- ☐ C tracheae and spiracles
- ☐ D book lungs and skin

157. RESPIRATION

MODERATE

SCRUTINISE CAREFULLY! TEST 2020 UNIT TEST 2

SANVI MANTRI

During hibernation, gaseous exchange in a frog occurs chiefly by:

- ☐ A pulmonary respiration
- ☐ B cutaneous respiration
- ☐ C branchial respiration
- ☐ D tracheal respiration

158. CIRCULATORY SYSTEM

EASY

The frog has a:

- ☐ A two-chambered heart
- ☐ B three-chambered heart
- ☐ C four-chambered heart
- ☐ D single-chambered heart

159. HEART

EASY

Oxygenated blood from the lungs and skin enters the:

- ☐ A right atrium
- ☐ B left atrium
- ☐ C ventricle directly
- ☐ D sinus venosus

160. HEART

EASY

The sinus venosus opens into the:

- ☐ A left atrium
- ☐ B right atrium
- ☐ C ventricle
- ☐ D truncus arteriosus

161. HEART

MODERATE

The chamber from which the truncus arteriosus arises is the:

- ☐ A ventricle
- ☐ B left atrium
- ☐ C right atrium
- ☐ D sinus venosus

162. BLOOD

EASY

Compared with mature human red blood cells, frog red blood cells are typically:

- ☐ A smaller and without membrane
- ☐ B without haemoglobin
- ☐ C nucleated and oval
- ☐ D fragmented into platelets

163. CIRCULATION

MODERATE

Circulation in a frog is called incomplete double circulation because:

- ☐ A blood passes through the heart twice, with some mixing in the single ventricle
- ☐ B only lymph circulates twice
- ☐ C the heart has four completely separated chambers
- ☐ D blood passes through the heart once and never mixes

164. PORTAL SYSTEMS

MODERATE

SCRUTINIZING NEET 2028 UNIT TEST 3

SANVI MANTRI

Frogs possess:

- ☐ A only a hepatic portal system
- ☐ B both hepatic and renal portal systems
- ☐ C only a renal portal system
- ☐ D no portal system

165. LYMPHATIC SYSTEM

EASY

Frog lymph differs from blood mainly because lymph:

- ☐ A contains no water
- ☐ B has abundant erythrocytes
- ☐ C lacks red blood cells and contains lymphocytes
- ☐ D flows only inside arteries

166. EXCRETORY SYSTEM

EASY

The chief nitrogenous waste excreted by an adult frog is:

- ☐ A ammonia
- ☐ B uric acid
- ☐ C urea
- ☐ D creatinine only

167. EXCRETORY SYSTEM

EASY

The kidneys of an adult frog are:

- ☐ A single and located inside the liver
- ☐ B paired, elongated and attached to the dorsal body wall
- ☐ C paired and suspended from the lungs
- ☐ D absent in adults

168. EXCRETORY SYSTEM

MODERATE

The structural and functional excretory units in a frog kidney are:

- ☐ A alveoli
- ☐ B Malpighian tubules
- ☐ C nephrons or uriniferous tubules
- ☐ D flame cells

169. URINOGENITAL SYSTEM

MODERATE

In a male frog, the ureters also function as:

- ☐ A oviducts
- ☐ B hepatic portal veins
- ☐ C bile ducts
- ☐ D urinogenital ducts

170. URINOGENITAL SYSTEM

MODERATE

Which statement about a female frog is correct?

- ☐ A The ureters carry both ova and urine
- ☐ B Oviducts and ureters open separately into the cloaca
- ☐ C Ovaries are directly connected to kidneys by vasa efferentia
- ☐ D The urinary bladder is absent

171.	NERVOUS SYSTEM	EASY
SCRUTINY 2020-21 CBSE T2020 UNIT TEST 2		
The frog's central nervous system consists of the:		
A brain and sense organs only B cranial nerves only C brain and spinal cord D sympathetic chains only		
172.	BRAIN	EASY
The brain of a frog is divided into:		
A forebrain, midbrain and hindbrain B cerebrum and spinal cord only C thorax, abdomen and pelvis D sympathetic and parasympathetic lobes		
173.	CRANIAL NERVES	EASY
The number of pairs of cranial nerves in a frog is:		
A 10 B 12 C 8 D 31		
174.	SENSE ORGANS	MODERATE
Which structure is involved in hearing and equilibrium in a frog?		
A Nictitating membrane B Copulatory pad C Vocal sac D Internal ear		
175.	MALE REPRODUCTIVE SYSTEM	EASY
The testes of a male frog are attached near the kidneys by a fold called:		
A mesovarium B mesorchium C mesentery D pericardium		
176.	MALE REPRODUCTIVE SYSTEM	MODERATE
Sperm from each testis reaches the kidney through several:		
A ureters B hepatic ducts C oviducts D vasa efferentia		
177.	FEMALE REPRODUCTIVE SYSTEM	MODERATE
The funnel-shaped opening of each frog oviduct is located:		
A at the tip of the tongue B near the ovary in the body cavity C within the urinary bladder D inside the testis		
178.	REPRODUCTION	EASY
Fertilisation in frogs is generally:		
A internal in the uterus B internal in the cloaca C external in water D absent because development is direct		

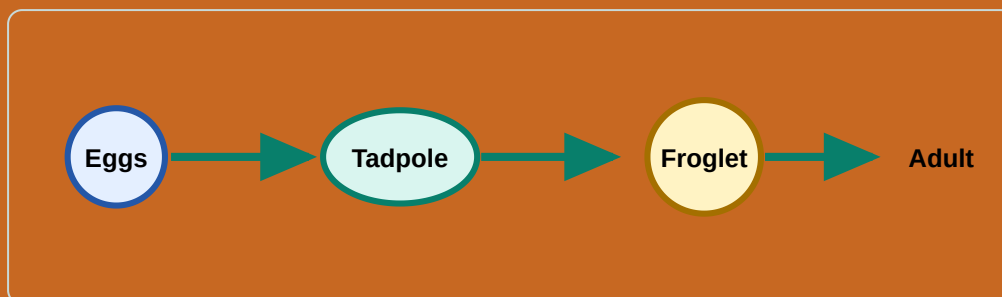
179.

LIFE CYCLE DIAGRAM

SCRUTINY ACADEMY • NEET 2028 UNIT TEST 3

EASY
SAINVI MANTRI

The diagram represents the normal developmental sequence in a frog.



The conversion of tadpole into froglet is called:

- ☐ A metamorphosis
 ☐ B budding
 ☐ C binary fission
 ☐ D regeneration

180.

ORGAN-SYSTEM INTEGRATION

MODERATE

Match each structure with its function.

Structure	Function
A. Nictitating membrane	1. Carries urine and sperm in male
B. Tympanum	2. Protects eye
C. Ureter	3. Receives sound vibration
D. Cloaca	4. Common terminal chamber

Select the correct matching.

- ☐ A A-2, B-3, C-1, D-4
 ☐ B A-3, B-2, C-4, D-1
 ☐ C A-2, B-4, C-3, D-1
 ☐ D A-1, B-3, C-2, D-4

ROUGH RESPONSE RECORD

Answer Sheet

Shade or circle one response for each question. Final scoring is performed by the online examination.

Physics					Chemistry					Botany					Zoology				
1	☐	☐	☐	☐	46	☐	☐	☐	☐	91	☐	☐	☐	☐	136	☐	☐	☐	☐
2	☐	☐	☐	☐	47	☐	☐	☐	☐	92	☐	☐	☐	☐	137	☐	☐	☐	☐
3	☐	☐	☐	☐	48	☐	☐	☐	☐	93	☐	☐	☐	☐	138	☐	☐	☐	☐
4	☐	☐	☐	☐	49	☐	☐	☐	☐	94	☐	☐	☐	☐	139	☐	☐	☐	☐
5	☐	☐	☐	☐	50	☐	☐	☐	☐	95	☐	☐	☐	☐	140	☐	☐	☐	☐
6	☐	☐	☐	☐	51	☐	☐	☐	☐	96	☐	☐	☐	☐	141	☐	☐	☐	☐
7	☐	☐	☐	☐	52	☐	☐	☐	☐	97	☐	☐	☐	☐	142	☐	☐	☐	☐
8	☐	☐	☐	☐	53	☐	☐	☐	☐	98	☐	☐	☐	☐	143	☐	☐	☐	☐
9	☐	☐	☐	☐	54	☐	☐	☐	☐	99	☐	☐	☐	☐	144	☐	☐	☐	☐
10	☐	☐	☐	☐	55	☐	☐	☐	☐	100	☐	☐	☐	☐	145	☐	☐	☐	☐
11	☐	☐	☐	☐	56	☐	☐	☐	☐	101	☐	☐	☐	☐	146	☐	☐	☐	☐

Physics					Chemistry					Botany					Zoology				
12	☐	☐	☐	☐	57	☐	☐	☐	☐	102	☐	☐	☐	☐	147	☐	☐	☐	☐
13	☐	☐	☐	☐	58	☐	☐	☐	☐	103	☐	☐	☐	☐	148	☐	☐	☐	☐
14	☐	☐	☐	☐	59	☐	☐	☐	☐	104	☐	☐	☐	☐	149	☐	☐	☐	☐
15	☐	☐	☐	☐	60	☐	☐	☐	☐	105	☐	☐	☐	☐	150	☐	☐	☐	☐
16	☐	☐	☐	☐	61	☐	☐	☐	☐	106	☐	☐	☐	☐	151	☐	☐	☐	☐
17	☐	☐	☐	☐	62	☐	☐	☐	☐	107	☐	☐	☐	☐	152	☐	☐	☐	☐
18	☐	☐	☐	☐	63	☐	☐	☐	☐	108	☐	☐	☐	☐	153	☐	☐	☐	☐
19	☐	☐	☐	☐	64	☐	☐	☐	☐	109	☐	☐	☐	☐	154	☐	☐	☐	☐
20	☐	☐	☐	☐	65	☐	☐	☐	☐	110	☐	☐	☐	☐	155	☐	☐	☐	☐
21	☐	☐	☐	☐	66	☐	☐	☐	☐	111	☐	☐	☐	☐	156	☐	☐	☐	☐
22	☐	☐	☐	☐	67	☐	☐	☐	☐	112	☐	☐	☐	☐	157	☐	☐	☐	☐
23	☐	☐	☐	☐	68	☐	☐	☐	☐	113	☐	☐	☐	☐	158	☐	☐	☐	☐
24	☐	☐	☐	☐	69	☐	☐	☐	☐	114	☐	☐	☐	☐	159	☐	☐	☐	☐
25	☐	☐	☐	☐	70	☐	☐	☐	☐	115	☐	☐	☐	☐	160	☐	☐	☐	☐
26	☐	☐	☐	☐	71	☐	☐	☐	☐	116	☐	☐	☐	☐	161	☐	☐	☐	☐
27	☐	☐	☐	☐	72	☐	☐	☐	☐	117	☐	☐	☐	☐	162	☐	☐	☐	☐
28	☐	☐	☐	☐	73	☐	☐	☐	☐	118	☐	☐	☐	☐	163	☐	☐	☐	☐
29	☐	☐	☐	☐	74	☐	☐	☐	☐	119	☐	☐	☐	☐	164	☐	☐	☐	☐
30	☐	☐	☐	☐	75	☐	☐	☐	☐	120	☐	☐	☐	☐	165	☐	☐	☐	☐
31	☐	☐	☐	☐	76	☐	☐	☐	☐	121	☐	☐	☐	☐	166	☐	☐	☐	☐
32	☐	☐	☐	☐	77	☐	☐	☐	☐	122	☐	☐	☐	☐	167	☐	☐	☐	☐
33	☐	☐	☐	☐	78	☐	☐	☐	☐	123	☐	☐	☐	☐	168	☐	☐	☐	☐
34	☐	☐	☐	☐	79	☐	☐	☐	☐	124	☐	☐	☐	☐	169	☐	☐	☐	☐
35	☐	☐	☐	☐	80	☐	☐	☐	☐	125	☐	☐	☐	☐	170	☐	☐	☐	☐
36	☐	☐	☐	☐	81	☐	☐	☐	☐	126	☐	☐	☐	☐	171	☐	☐	☐	☐
37	☐	☐	☐	☐	82	☐	☐	☐	☐	127	☐	☐	☐	☐	172	☐	☐	☐	☐
38	☐	☐	☐	☐	83	☐	☐	☐	☐	128	☐	☐	☐	☐	173	☐	☐	☐	☐
39	☐	☐	☐	☐	84	☐	☐	☐	☐	129	☐	☐	☐	☐	174	☐	☐	☐	☐
40	☐	☐	☐	☐	85	☐	☐	☐	☐	130	☐	☐	☐	☐	175	☐	☐	☐	☐
41	☐	☐	☐	☐	86	☐	☐	☐	☐	131	☐	☐	☐	☐	176	☐	☐	☐	☐
42	☐	☐	☐	☐	87	☐	☐	☐	☐	132	☐	☐	☐	☐	177	☐	☐	☐	☐
43	☐	☐	☐	☐	88	☐	☐	☐	☐	133	☐	☐	☐	☐	178	☐	☐	☐	☐
44	☐	☐	☐	☐	89	☐	☐	☐	☐	134	☐	☐	☐	☐	179	☐	☐	☐	☐
45	☐	☐	☐	☐	90	☐	☐	☐	☐	135	☐	☐	☐	☐	180	☐	☐	☐	☐

Correct: _____ × 4 = _____

Incorrect: _____ × 1 = _____

Final score: _____ / 720